

Lognormal regression – three views

1. Index form

The per-observation reading: what happens for each row i of your data.

$$\begin{aligned} y_i \mid \mu_i, \sigma_i &\sim \text{Lognormal}(\mu_i, \sigma_i^2) &\iff & \log(y_i) \mid \mu_i, \sigma_i \sim \text{Normal}(\mu_i, \sigma_i^2) \\ \mu_i &= \beta_0 + \beta_1 x_i \\ \log(\sigma_i) &= \gamma_0 \end{aligned}$$

2. Matrix form

The same model in matrix notation. The structural contract every textbook past chapter 4 switches to.

$$\begin{aligned} \log(\mathbf{y}) \mid \boldsymbol{\mu}, \boldsymbol{\sigma} &\sim \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \\ \boldsymbol{\mu} &= \mathbf{X}\boldsymbol{\beta} \\ \log(\boldsymbol{\sigma}) &= \mathbf{Z}\boldsymbol{\gamma} \end{aligned}$$

3. Worked observation ($i = 1$)

The index-form equations evaluated at the first observation of your data, with coefficient estimates plugged in.

$$\underbrace{y_1}_{\text{observed}=4.78} = 2.02 + -0.0136 \cdot 0.409 + 2.77 = 4.78$$

Predicted $\hat{\mu}_1 = 2.01$; residual $\hat{\varepsilon}_1 = 2.77$.

And the σ submodel at $i = 1$:

$$\log \hat{\sigma}_1 = -0.764 \Rightarrow \hat{\sigma}_1 = \exp(-0.764) \approx 0.466$$